

Chapter 83

The Long Game

What the book actually costs in hours, the pace that survives a bad month, how to choose your own summit, and the four plateaus that arrive after the first job.

Draft chapter · v1

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You will learn to: see the real arithmetic of this book, in hours and in years, and plan against it rather than against a feeling · choose a weekly pace that survives a bad month · tell the difference between reading that helps and reading that hides · build depth in one place and literacy everywhere else · pick your own stopping point on the career tree and stop there without apology · separate what will still be true in ten years from what will not · find the feedback that no book can supply · recognize the four plateaus that arrive after the first job, which look nothing like the first one · keep the only skill that never goes out of date.

Before you start: the book. Chapter 9 in particular, which this chapter is the far end of.

Time needed: about an hour to read. The rest of it takes years, which is the subject.

Tools: none.

Practice data: none, except your own last week, which you will be asked to count truthfully.

A note on the shape of this chapter. There are no exercises and no answer key. There is one week's worth of work at the end, and six decisions that only you can make. This is the last chapter, and the last thing it should do is give you something else to read.

Why this matters

Eighty-two chapters ago this book made a promise: that the whole territory of a data career could be laid out in one place, from a first query against one table to the design of systems an organization depends on.

That promise has now been kept, and keeping it creates a new problem. **You can see the whole map, and you have walked almost none of it.** The map took you a few hundred hours to read. The territory takes years to cross. Everything this book taught you about the difference between knowing and doing applies most sharply to the book itself.

So the last chapter is not about data. It is about the distance between here and the version of you who does this for a living, and about the small number of habits that decide whether you cover that distance or stop in the foothills.

Most people stop in the foothills. Not because the material was too hard. Because nobody told them how long it takes, so they measured themselves against an imaginary schedule and concluded they were failing.

In plain English

This book is a map of a mountain, and you have now finished reading the map.

A map is an extraordinary thing to have. It shows the routes, the false summits, where the paths join, which ridges lead nowhere, and roughly how long each stage takes. Somebody who has the map is in a completely different position from somebody standing at the bottom looking up.

But nobody has ever arrived at a summit by reading. The map tells you the climb takes four days; it cannot make your legs do four days of walking. And the most common way to fail is not to take a wrong path. It is to look at the height of the mountain, compare it to the distance you covered today, and decide the whole thing is unreasonable.

Everything in this chapter is about walking: the pace that works, the weight to carry, what to do on the days when nothing moves, and how to choose which summit is actually yours.

83.1 The arithmetic, stated plainly

Chapter 9 said the honest thing about time: months to become a job-ready analyst, years to go further, and no course compresses that. Now that the book exists, the estimate can be exact, because every chapter carries its own.

Adding up the *Time needed* line of all sixty-seven teaching chapters gives **764 to 980 hours**.

	Chapters	Hours	At 6 hours a week
Parts 0 and I, the foundations	1-9	29-39	5-6 weeks
Part II, The Analyst	10-27	290-357	48-60 weeks
Parts 0 to II: job-ready	1-27	319-396	12 to 15 months
Part III, advanced analytics	28-34	92-122	15-20 weeks
Part IV, machine learning	35-44	86-121	14-20 weeks
Part V, data engineering	45-52	100-132	17-22 weeks
Part VI, production ML and GenAI	53-59	91-116	15-19 weeks
Part VII, architecture and leadership	60-67	76-93	13-16 weeks
All of it	1-67	764-980	2.4 to 3.1 years

What this book actually costs, in hours

Added up from the Time needed line of all sixty-seven teaching chapters. Bars show the low and high estimate.

Part		Hours	At 6 hours a week
Parts 0 and I	Ch 1–9	29–39	5–6 weeks
Part II, The Analyst	Ch 10–27	290–357	48–60 weeks
Part III	Ch 28–34	92–122	15–20 weeks
Part IV	Ch 35–44	86–121	14–20 weeks
Part V	Ch 45–52	100–132	17–22 weeks
Part VI	Ch 53–59	91–116	15–19 weeks
Part VII	Ch 60–67	76–93	13–16 weeks
Job-ready: Parts 0-II	Ch 1–27	319–396 hours	12 to 15 months
The whole map	Ch 1–67	764–980 hours	2.4 to 3.1 years

Being hireable is a quarter of the book. That is the number nobody is given, and it is the one you can plan against.

Figure 83.1 – Being hireable is a quarter of the book. The other three quarters are the rest of a career, and they are optional.

Read the bold rows first. **The path to being hireable as an analyst is about 320 to 400 hours**, which at six hours a week, a realistic figure alongside a job, is a little over a year. The whole book, done properly, is two and a half to three years of part-time evenings.

Three things follow from that table, and they are worth more than any encouragement.

The number is not discouraging; it is a plan. "A year or so" is something you can arrange your life around. "However long it takes" is not, and it is what most people are working to, which is why they cannot tell whether they are on schedule and eventually assume they are not.

Job-ready is a quarter of the book. Parts 0 to II are 319 to 396 hours of the 764 to 980. You do not need the other three quarters to be hired, and Chapter 8's career tree exists precisely so you can see how far your own branch actually goes.

These are reading-and-exercise hours, not fluency hours. Chapter 9 section 9.1 made this distinction and it matters here: finishing Chapters 12 and 13 means you can write the queries they teach. Being quick on a messy dataset you did not design takes more. Plan the chapter hours, then plan again.

83.2 Consistency beats intensity

This is the single most important thing to take from the chapter, and it is arithmetic rather than inspiration.

Consider two people who start this book on the same Monday.

The sprinter decides to do it properly: twenty-five hours a week, evenings and both weekend days, everything else on hold. This is impressive for a while, and it works. Call it twelve weeks, which is a generous estimate of how long that pace survives alongside a full-time job, and it ends the way it usually ends: a bad week at work, a missed weekend, a week off that becomes a month off, and a quiet decision not to go back. Total: **300 hours**.

The steady one gives six hours a week. Two evenings and a Saturday morning. It never feels like enough. Nothing dramatic happens in any single week.

	The sprinter	The steady one
Week 12	300 hours, and stops	72 hours
Week 50	300 hours	300 hours: the crossover
Week 66	300 hours	396 hours: finishes Parts 0 to II
Year 3	300 hours	936 hours: has done the whole book

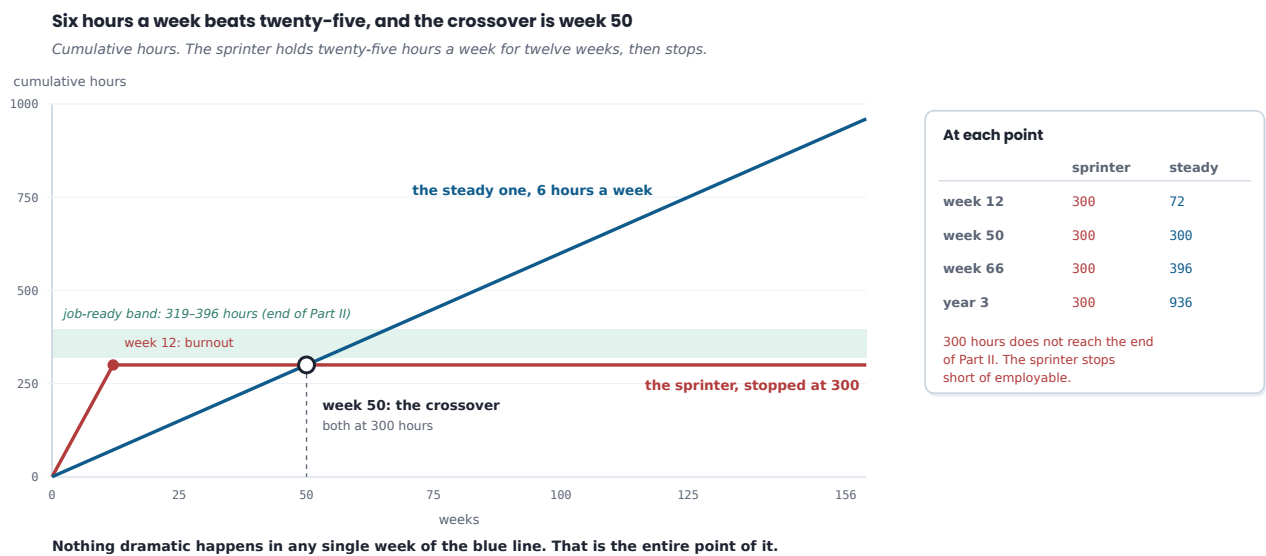


Figure 83.2 – The sprinter's line stops below the green band. That band is the difference between studying and being employable.

The steady one passes the sprinter in **week 50**, and after that it is not a race any more.

Now the part that should stop you. The sprinter's 300 hours do not reach the end of Part II, which needs 319 to 396 hours from the start of the book. **The sprinter burns out somewhere between one chapter and five chapters short of being employable**, depending on how fast they were actually absorbing it. They did four times the weekly work the steady one ever did, they have nothing anybody would hire them for, and the reason is not talent or discipline. It is that they chose a pace they could not hold, on a route that was longer than they had been told.

Progress here compounds in a specific, literal way. The SQL from Chapter 12 is used in Chapter 13, and in Chapter 14's cleaning, and in Chapter 16's model, and in Chapter 28's window functions, and in Chapter 32's dbt models, and in Chapter 49's warehouse, and in every interview in Part VIII. An hour spent on foundations is not spent once. It is a deposit that every later chapter withdraws from.

So protect a modest weekly number and defend it. Six hours you actually do beats fifteen you intend. Put them in the calendar as appointments with a name, treat them the way you would treat a meeting with your manager, and when a week is lost, which it will be, return to the same six. The people who get there are rarely the most talented. They are the ones who did not quit, and quitting is almost always downstream of a pace that was never survivable.

83.3 Study just enough to build, then build

Chapter 9 section 9.3 made the argument: study produces knowledge, projects produce skill, and feedback over time produces judgment. People who stall are almost always oversupplied with the first.

Here is the version that matters after the book, because the failure mode changes shape.

While you were reading, the projects were handed to you at the end of every chapter. After the book, nobody hands you anything, and reading becomes available in unlimited quantities at any hour, always slightly easier than building and always feeling like progress. That is the trap, and it has a tell.

The tell: if you can describe a technique but cannot name a time you used it on data you did not design, you have knowledge and not skill. Go and find the data.

Three sources, in order of value:

1. **Your own work.** The highest-value practice data in the world is the messy export from the system your employer actually uses, answering a question somebody actually asked. Chapter 27's project is built this way for a reason. Check what you are allowed to use and what must never leave the building, which Chapter 64 covers.
2. **A real public dataset with a real question attached.** Government open data, a regulator's filings, your city's transport figures. The question is the hard part and the part that makes it a project. Anything from a competition site arrives pre-cleaned, which removes most of the work that the job consists of.
3. **A thing somebody needs.** A local business, a community organization, a friend running a shop. Unpaid, small, and worth more than six tutorials, because somebody will use the output and tell you when it is wrong.

And write it down where it can be seen. Chapter 26 gave you the repository; Chapter 27 gave you the standard. Work that nobody can look at cannot be reviewed, and unreviewed work is where bad habits go to become permanent.

83.4 Go deep, then broad

The strongest careers are **T-shaped**: real depth in one area, working literacy across the rest.

Depth is what makes you valuable, because depth is what nobody can hire quickly. Breadth is what lets you be useful outside your own lane: it is what makes you able to talk to the data engineer without a translator, to know when a question is a machine learning question and when it is a `GROUP BY`, and eventually to design something that spans more than one team.

The mistake is doing them in the wrong order. Breadth first produces someone who can name every tool and finish nothing, and it is a quick state to reach now that a survey of any field is one prompt away. Depth first is slower and less comfortable, because for a while you know a great deal about one thing and feel ignorant about everything else. That feeling is what depth costs.

Practically: pick one column of Chapter 8's career tree and go until you are the person others ask. Read the rest of this book for literacy, not mastery, and revisit those parts properly only when a real problem sends you there. A chapter you return to because you need it will teach you more in an evening than four read in sequence because they were next.

83.5 Choose your own summit

This book is named for the full climb, and it would be dishonest to leave you thinking you have to make it.

You do not. The career tree branches for a reason. A senior data analyst who understands a business deeply is enormously valuable, is paid well, and never has to touch Kubernetes. A superb data engineer need never train a model. A business analyst who can turn a vague ask into a specification that survives contact with a build is rarer than either, and Chapter 25 exists because of it.

Every tier on that tree is a real destination where people spend satisfying careers, not a checkpoint on the way to somewhere better. The point of seeing the whole map is not to force yourself to the top of it. It is to **choose your own summit with open eyes**, knowing what the others involve, and to know the routes in case you change your mind.

Two questions worth being honest about, because the answers do not change much over a career:

- **What do you want your Tuesday to look like?** Not your title. Your Tuesday. Architects spend it in conversations and documents. Analysts spend it in data and with the people who asked. Engineers spend it building and repairing things that run without them. These are different lives, and one of them suits you more than the others.
- **What are you willing to be on call for?** The further up the tree, the more of other people's failures become yours at inconvenient hours. Some people find that meaningful. Some find it corrosive. Both answers are correct and only one of them is yours.

There is no prize for reaching a summit you did not want.

83.6 The fundamentals are durable; the tools are not

This field moves quickly enough to make anyone feel permanently behind. There is always a newer framework, a better model, a tool that makes last year's tool look slow, and a stream of people announcing that everything you learned is now obsolete.

The defense against that anxiety is a discipline, not a mood: **invest most of your hours in the things that do not change.**

Durable, and worth years	Perishable, and worth an afternoon when you need it
What a row means, and what a join does to it	Whichever query tool your company installed
How a database decides to answer a query	This year's warehouse vendor

Durable, and worth years	Perishable, and worth an afternoon when you need it
What makes an experiment valid	The A/B testing platform's interface
Why a model's error is the number that matters	The particular library's function names
How to tell a plausible number from a right one	The dashboard tool's formula language
How to write so that a busy person acts correctly	The template your company uses this quarter
Reproducible work: version control, tests, a README	The specific CI service

The left column has lasted decades and will outlast whatever is trending when you read this. The right column is a series of interfaces over the left column, and once you know the left column properly, a new tool in the right one takes a weekend rather than a season. That is why the people who seem to pick up new tools effortlessly are usually not fast learners. They are people who already understood what the tool is for.

One caution against the opposite error. "Fundamentals are durable" is not a license to ignore what is happening. Something new in kind arrives every few years, and the assistants of Chapter 26 section 26.11 are the current one: they change what a working day contains, and pretending otherwise is its own kind of obsolescence. The test is whether a change alters *what the job is* or only *which buttons it uses*. Learn the first kind properly and promptly. Learn the second kind when it is in front of you.

83.7 You cannot do this alone

The third ingredient from Chapter 9, feedback and time, is the one no book can supply, and the fastest way to get it is other people.

Learning in isolation plateaus. Learning with feedback compounds. That is not encouragement, it is a description of a mechanism: without an outside reader you can only find the mistakes you already know how to look for, which by definition excludes the ones holding you back.

Four sources, from easiest to most valuable:

- **Communities.** A forum, a local meetup, a professional group, a study group of three. The bar is low: somewhere you can ask a question without being embarrassed and answer one without being an expert.
- **Code review.** Chapter 26 section 26.7 taught the mechanism. Ask for it specifically, on something small, with a question attached, because "any feedback welcome" usually receives none. The review you learn most from is one you asked a precise question in.
- **A mentor,** which Chapter 9 section 9.6 covers properly. Not a formal arrangement in most cases: somebody a few years ahead who will look at your work occasionally and tell you the truth about it.
- **Teaching.** The most efficient learning there is. Explaining a topic exposes every place you only half understand it, immediately and uncomfortably. Write the post, give the lunchtime talk, answer the question on the forum. You do not need to be an expert to teach someone one step behind you, and the person one step behind you is the one you can help most.

Put your work where it can be seen and let people tell you where it is weak. That is the whole of it, and most people avoid it for years because it is the only uncomfortable item on this list.

83.8 The plateaus that come later

Chapter 9 section 9.7 dealt with the learning plateau: the stretch where practice continues and the weekly check does not move. Farah's log showed the shape of it and what breaks it, and you will meet your first one long before you finish Part II.

The plateaus that arrive after the first job look nothing like that one, and the difference matters: **a learning plateau is solved by changing how you practice. A career plateau is solved by changing what you are responsible for.** More practice does not touch any of the four below, which is why people stuck in them so often respond by taking another course and feeling no better afterward.

The four, and what each one actually is

1. The competence plateau. You are good at your job. Everything asked of you, you can do. Nothing in a week takes you by surprise.

The tell: you have not been out of your depth in six months.

Why it is a plateau: there is no more technique to add at the size of problem you are given, so effort stops converting into growth. The missing ingredient is not skill, it is scope.

The move: take on something you do not know how to do, in front of people, with a deadline. The pipeline nobody owns. The question that spans two teams. Volunteering for work slightly beyond you is the only reliable way to be given work slightly beyond you.

2. The stack plateau. Your company's tools stopped changing. You are doing your third identical year.

The tell: your CV would read the same as it did three years ago, with different dates.

Why it is a plateau: you are accumulating repetitions of something you already learned. That is worth something, and it is not growth.

The move: change the problem rather than the employer, at least first. Every company has a migration nobody wants, a source system nobody has modeled, a report that breaks monthly and is nobody's job. Those are unclaimed scope, and claiming one is faster than changing jobs and usually teaches more.

3. The indispensable plateau. You are the person everyone asks. It is flattering, it is useful to everyone around you, and it is quietly the most costly one on this list.

The tell: your calendar is other people's questions, and the work you would call yours happens after six.

Why it is a plateau: answering is not building. Being the bottleneck feels like being valuable and is in fact a cap, because your ceiling becomes the number of questions you can personally answer in a week.

The move: the whole of Part II is the answer, applied to yourself. Write it down (Chapter 26), automate it (Chapter 20), give it a self-service model somebody else can query (Chapter 16), and train a successor. Every one of those converts a question you answer forever into an answer that exists without you.

4. The ladder plateau. You keep getting better and nothing changes, because the next step up is not more of this job. It is a different job.

The tell: you are clearly stronger than you were two years ago, and your work, title and scope are identical.

Why it is a plateau: above a certain point, progress stops coming from being better at the craft and starts coming from leverage: designing what others build, or leading the people who build it. Chapter 67 describes that shift without softening it, including the part where your own keyboard stops being where your impact comes from.

The move: decide, deliberately, whether you want that. Many excellent people do not, and staying is a legitimate choice, not a failure, as long as it is a choice. What does not work is wanting the next rung and continuing to compete for it on craft alone.

The one that is not a plateau

Sometimes the flat stretch is not a plateau at all. It is a job you have outgrown, a manager who will not give you scope, or a company where the data work is decorative. Those do not resolve by trying harder either, and the honest signal is this: **if you can name the specific, reasonable thing you need in order to grow, and you have asked for it twice, and the answer keeps being not now, the constraint is not you.** Part VIII exists for that moment.

83.9 The meta-skill: learning how to learn

Every specific thing in this book will eventually age. The SQL will last longest and even it will change. The tools of Parts V and VI will change fastest; some of them will have been renamed before this book is a year old.

What does not age is the ability to teach yourself the next thing: to read documentation and build understanding from it, to construct a small experiment that tells you whether you have understood, to notice the moment you started nodding along without following, and to stay curious rather than defensive when the ground moves.

You have been practicing that skill for the whole book without it being named. Every chapter's structure was a rehearsal of it: a plain-English model first, then the mechanism, then a worked example you could check, then a project where nothing matched the example, then a set of questions to find out whether you actually had it. That sequence is not a book format. It is how to learn anything technical, and you can run it on a subject nobody has written a chapter about.

The portable version, for the next thing you have to learn:

1. Get one small, real thing working end to end before you understand it. Understanding follows a working example far more reliably than it precedes one.
2. Then go back and find out why it worked, line by line, until you could rebuild it without looking.
3. Change one setting deliberately and predict what will happen before you run it. Being wrong is the fastest part of the process.
4. Use it on a real question of your own, where nothing matches the tutorial.
5. Explain it to somebody, and notice exactly where you go vague. That is where the hole is.

If this book leaves you a more capable, self-directed learner, it has done its most important job, whatever tools you end up using.

83.10 What this book could not give you

It is worth being exact about this, because the gap is where the next few years go.

It could not give you judgment. Chapter 9 called judgment the product of skill plus feedback plus time, and there is no shortcut in the ingredient list. Judgment is knowing which of two correct approaches is right *here*, and it is assembled from having been wrong in specific, memorable ways. This book handed you other people's mistakes, which helps and is not the same thing.

It could not give you a domain. Every number in this book came from a company that does not exist. Real analysis is inseparable from knowing what a cancelled order means at *your* company, who overrides the discount, and which system is lying. That knowledge takes a year at a place and does not transfer whole.

It could not give you stakes. Nothing you did in these chapters mattered to anybody. The first time a decision is made on your number, you will check it differently, and that reflex cannot be installed by reading.

It could not give you the room. The meetings where things are actually decided, the argument about which figure is right, the colleague who explains in four minutes what a chapter took an hour to say. You get into that room by doing the work in front of people, and only by that.

None of this is a disappointment. It is the reason the field is worth a career rather than a weekend: the part that cannot be written down is the part you get paid for.

The ways people fall off the path

The move	What it looks like	The correction
Choosing a pace you cannot hold	twenty hours a week for a month, then nothing for three	pick a weekly number you could keep during a bad month, and defend it
Measuring against an imaginary schedule	"everyone else learned this faster"	measure against section 83.1's hours and your own honest weekly budget

The move	What it looks like	The correction
Reading instead of building	many finished courses, nothing anyone can open	study enough to start, then build. The tell is in section 83.3
Restarting	a fourth beginner course in a new tool	the discomfort of the intermediate stage is the work, not a sign of a wrong choice
Chasing tools	learning whatever is new rather than whatever is durable	the left column of section 83.6 first, always
Learning alone	no reviewer, no reader, no question ever asked	one of the four sources in section 83.7, this month
Waiting to feel ready	"one more chapter and then I'll apply"	Chapter 68 says what hiring actually tests. It is not readiness
Treating a career plateau as a learning plateau	another course, no change in the work	section 83.8: change what you are responsible for
Confusing being busy with being valuable	the calendar is full of other people's questions	the indispensable plateau, and the four fixes in section 83.8
Climbing a mountain you did not choose	pursuing a title whose Tuesday you would dislike	section 83.5's two questions, answered honestly
Quitting during the worst month	a decision made in a bad week	decide about a career in a good month, never a bad week

In the real world: how Meera got there

Meera Iyer is the person everyone at Riverstone asks for numbers. It is worth knowing how recent that is.

She did not study data. She joined in an administrative role, and when Imran left she inherited the Friday file from Chapter 2: one spreadsheet, on one laptop, that nobody else could update. She kept it running because somebody had to.

What she did next was ordinary and is the whole point. She learned enough SQL to stop exporting the same report by hand. Not on a course and not in a burst: an hour most mornings before the office filled up, and a longer stretch on Saturdays when she could manage it. There were months with nothing in them. The file kept running throughout, which is what made the practice stick, because every query she learned had something to do the same week.

Somewhere in the second year the questions started coming to her instead of to the system, and by the third she was the person the sales head checked a number with before a board meeting. Nobody appointed her. The gap closed on its own.

Then she hit the third plateau in section 83.8, and did not recognize it as one for most of a year.

By 2025 her mornings had a shape: forty minutes building the Daily Sales Flash by hand, then most of Friday on the sales pack, then copying overdue invoices out of the finance system into emails for the reps. She was busier than she had ever been and had not learned anything new in months. When

Anita asked what she needed, she asked for more time, which was the wrong answer and the one that felt natural.

What changed it was not a course. It was writing down what the forty minutes actually consisted of, and noticing that all of it was a query she already knew how to write and a template she could have built in an afternoon. Chapter 20's automated flash is that afternoon. The Power Query rebuild of Vikram's month-end pack in Chapter 11 is another. Each one converted a thing she did forever into a thing that happened without her, and the hours came back as hours she could spend on work she had never done before.

The job posting in Chapter 8, *Data Analyst (Sales Analytics and Automation)*, exists because of that pattern. Anita wrote it with her. The word **Automation** is in the title because Meera had learned, expensively and slowly, that being the bottleneck is not the same as being valuable.

Five years, an hour most mornings, on real work that mattered to somebody that week. There is no other version of this story, and it does not require anybody to have been remarkable.

The first week after this book

Not a project. One week, and then you are on your own.

Monday. Count it. Write down the hours you gave to learning last week, truthfully, not the hours you meant to. That number, not an aspiration, is what you plan with. Compare it against section 83.1's table and write down the date you would reach the end of Part II at that pace. Put the date somewhere you will see it.

Tuesday. Put the hours in the calendar. As named appointments, in the slots you actually have, for the next twelve weeks. Protect them the way you would protect a meeting with your manager, because that is what they are.

Wednesday. Choose the question. One real question, from your own work if you are allowed, that somebody would care about the answer to. Chapter 27 section 27.1's three tests. Write it as a sentence a named person would say.

Thursday. Make it public. A repository, a README with the question in the first paragraph (Chapter 26 section 26.9), and a first commit. Empty is fine. The point is that it exists and has your name on it.

Friday. Find the one person. One place you will get feedback: a community, a study group, a colleague who will review something, somebody a year ahead. Section 83.7. Make contact this week, not when you have something impressive.

Saturday. Do two hours on the question. Not on reading about the question.

Sunday. Decide your summit. Answer section 83.5's two questions in writing: what you want your Tuesday to look like, and what you are willing to be on call for. Then look at Chapter 8's tree and mark the branch. You can change it later. You cannot aim without it.

How you will know it is working

- You have a weekly number of hours, it is written down, and it survived a bad week in the last month.
 - You can name the date you expect to finish your current part, and it came from arithmetic rather than hope.
 - There is something with your name on it that another person could open today.
 - You have been out of your depth on purpose in the last three months.
 - At least one person who is not you has looked at your work and said something uncomfortable about it.
 - You can tell someone which branch of the career tree you are on and why you chose it.
 - When something new appears, you can say whether it changes the job or only the buttons.
 - You have stopped counting courses.
 - When you go flat, you check which kind of plateau it is before deciding what to do about it.
 - You have taught somebody one step behind you at least one thing.
-

Recap

The book is 764 to 980 hours across sixty-seven teaching chapters, and 319 to 396 of those reach the end of Part II, which is the point at which people get hired. At six hours a week, job-ready is about twelve to fifteen months and the whole map is two and a half to three years. That is a plan you can arrange a life around, and it is the number most people never get told.

Consistency beats intensity, arithmetically. Six hours a week passes twenty-five-hours-a-week-for-twelve-weeks in week 50, and the sprinter runs out roughly one chapter short of employable. Choose a pace that survives a bad month and defend it.

Study enough to start building, then build, because after the book nobody hands you projects and reading is always available and always feels like progress. Go deep in one branch, keep literacy across the rest, and choose your own summit: every tier on the tree is a destination where people spend good careers, and there is no prize for reaching one you did not want.

Put your hours into the durable column, which is what a row means, what a join does, why an experiment is valid, how to tell a plausible number from a right one. Tools sit on top of that and take a weekend once the foundation is there. Stay alert to the rare change that alters what the job is, rather than which buttons it uses.

You cannot do this alone, and learning with feedback compounds while learning in isolation plateaus. Communities, code review, a mentor, and teaching, which is the most efficient learning there is.

The plateaus after the first job are different in kind from Chapter 9's: they are solved by changing what you are responsible for, not how you practice. Competence, stack, indispensable, ladder. The indispensable one is the most comfortable and the most costly.

The one thing that never ages is learning how to learn, and the whole book was a rehearsal of it: working example first, then the mechanism, then a deliberate change with a prediction, then your own question, then explaining it to somebody.

And what a book cannot give you is judgment, a domain, stakes, and the room where things are decided. That gap is not a flaw in the book. It is the reason this is worth a career.

Six decisions, which only you can make

No answers, and there will not be any.

1. How many hours a week, truthfully, and which hours specifically?
 2. Which branch of Chapter 8's tree, for now?
 3. What is the first real question you are going to answer, and who would care about the answer?
 4. Who is going to tell you the truth about your work?
 5. What is the thing you are currently the bottleneck for, and what would it take to not be?
 6. What will you have to show for the next twelve months, and how will you know in month six whether you are on track?
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Key terms

compounding · consistency over intensity · weekly budget · fluency hours · T-shaped · depth and breadth · career tree · your own summit · durable fundamentals · perishable tools · feedback loop · mentorship · teaching as learning · learning plateau (Chapter 9) · competence plateau · stack plateau · indispensable plateau · ladder plateau · scope · leverage · learning how to learn · judgment

(All terms are defined in the Glossary, Appendix A.)

Where this leads

Nowhere in this book. That was the last chapter.

If you are preparing to be hired, Part VIII, Chapters 68 to 82, is the part that is written to be used rather than read: how hiring actually works, the question banks for every track, the behavioral chapter, and full take-homes with model submissions. Use it as a reference during a search, not as more reading now.

Everything else is outside these pages: the question somebody asks you on Thursday, the number you have to defend, the system you eventually get to design, and the first person you mentor who does for somebody else what this book has tried to do for you.

And so we arrive where every honest guide has to end.

The map is not the territory. You now know the terrain, the routes, and the destinations. But knowing the path and walking it are different things, and only one of them changes anything.

The reading is finished. The building begins. Open a database, load some real data, and write the first query of the rest of your career.

The mountain is real. You have the map.

Go climb.